How Reproducible Is a Guard Evaluation? A Measured Floor, and Where It Isn't Small

Vikram Jha

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> ***Preprint.*** *This is the named version of a paper prepared for peer review. The anonymized submission build is identical in content; only the author block, the companion-paper citation keys and the anonymization note differ.*

> ***Pre-registered before any measurement.*** *Hypotheses, arms, the acceptability comparison and the invalidation conditions were frozen and hashed (`5435ed47`) before the first run executed.* ***Two of the three hypotheses were refuted, including the one that motivated the paper.*** *Both refutations are reported in place.*

Abstract

Guard models are compared, ranked and selected on the basis of evaluation runs whose *reproducibility has never been measured.* We ran the same guard, on the same corpus, at the same quantization, seed, temperature and context size, eighteen times, and measured how often it returns a different verdict.

The floor is small: 998 of 1,000 items returned identical verdicts across eight independent runs, for a mean pairwise disagreement of **0.09%**. Differences above roughly 0.1% between separately-run guard evaluations are therefore real, not noise.

That result refutes the hypothesis that motivated the experiment. We predicted the floor would be large enough to swallow effects routinely published as findings — specifically our own prior measurement that `Q4_K_M` quantization changes 4 to 11 verdicts per thousand. It does not: those effects sit 4–12× above the floor and survive. **We also refute the standard folk explanation** for guard nondeterminism: varying thread count, which the serving stack's own documentation identifies as the culprit, changes nothing measurable.

The paper's remaining contribution is a discrepancy we can localize but not yet resolve. A prior experiment observed **1.0% and 5.7%** verdict drift under the same nominal configuration — ten to sixty times this floor. The two measurements differ in their *item population*: the high-drift sets were composed entirely of items sitting at a decision boundary by construction, while the corpus here is stratified and mostly not. **If the floor is population-dependent, then a single global floor is the wrong object**, and the number that matters — the floor on borderline items, which is precisely where effects are measured — remains unmeasured. We state this as an open hazard rather than absorbing it into a headline, and we release the protocol and the estimator so that any guard evaluation can report its own floor alongside its results.

Keywords: guard models, reproducibility, evaluation methodology, serving nondeterminism

1. Introduction

A guard model is a classifier deployed inline in an agentic pipeline, judging whether content is safe. Guards are compared against one another constantly — in vendor documentation, in benchmark tables, in procurement decisions — and those comparisons rest on evaluation runs.

An evaluation run is a measurement, and measurements have noise. For guard evaluation, the magnitude of that noise has, to our reading, never been reported. Papers state the model, usually the corpus, sometimes the temperature and seed. They do not state, because nobody has measured, how much the same evaluation would move if simply run again.

This matters because the effects being reported are often small. A guard comparison that finds one model two points better than another is making a claim about a difference; if two runs of the *same* model differ by two points, the claim is empty. The question is not rhetorical — it is arithmetic, and it has an answer.

1.1 What we expected, and what we found

We pre-registered three hypotheses. **Two were refuted.**

	hypothesis	outcome
H1	Agreement between two runs at identical configuration is below 1.0	**supported**, but barely
H2	The drift is attributable to thread count	**refuted**
H3	The floor is large enough to swallow effects published as findings	**refuted**

H3 was the paper's motivation. We expected to be able to say that a share of published guard comparisons report differences smaller than their own measurement noise. **The data does not support that claim, and we do not make it.**

1.2 Contributions

1. **The first measurement, to our knowledge, of guard-evaluation reproducibility**: 998/1,000 items unanimous across eight independent runs, 0.09% mean pairwise disagreement (§4).
2. **A refutation of the standard explanation.** Thread count, which the serving stack's own determinism note identifies as the source of nondeterminism, explains none of the observed drift (§4.3).
3. **A localized open hazard.** A 10–60× discrepancy against a prior measurement, attributable to item population, which implies the floor is not a single number (§5).
4. **A protocol and estimator** that lets any guard evaluation report its own floor for the cost of one additional run (§6).

2. Related work and positioning

Nondeterminism in LLM serving is documented at the systems level: batching, kernel selection, reduction order and thread count can all change floating-point results, and quantized inference compounds this. That literature establishes that variation *can* occur. It does not measure what it costs a downstream safety decision, which is the gap here.

Guard model evaluation is largely benchmark-aggregate. In our survey it reports model, corpus and metric, and does not report serving configuration beyond the model tag, does not repeat runs, and does not state a noise floor. We are not aware of prior work that measures repeated-run agreement for guard classifiers specifically.

Guard-model instability under perturbation is established [GuardMeaning], including the observation that flips concentrate in the ambiguous score region. That work perturbs the *input*; we hold the input fixed and perturb nothing at all, which is the complementary measurement and, to our reading, unreported.

Reproducibility work in ML has concentrated on training reproducibility — seeds, data order, hardware. Inference-time reproducibility of a *fixed* checkpoint has received far less attention, presumably because it is assumed to be exact. For quantized guards served through llama.cpp, it is not exact, and §4 measures by how much.

What we do not claim. We do not claim guard evaluations are unreliable — the measured floor says the opposite for the population we tested. We do not claim novelty on the existence of serving nondeterminism. Our contribution is its magnitude, its non-attribution to the standard cause, and the population-dependence hazard in §5.

3. Method

3.1 Design

One model, one corpus, one nominal configuration, repeated. Qwen3Guard-Gen-4B at `Q4_K_M`, 1,000 items stratified on the harm label from a public corpus, `num_ctx` 8192, seed 0, temperature 0, fail-closed scoring. Three arms:

arm	runs	what varies	isolates
A	8	**nothing declared**; separate containers	the floor a practitioner hits without changing anything
B	6	`num_thread` ∈ {2, 4, 8}, two runs each	whether thread count explains drift
C	4	A10G and L4, two runs each	whether GPU type contributes

Arm A is the headline, and its design principle is that nothing varies. Everything a practitioner would think to report — model, quantization, corpus, seed, temperature, context size — is fixed and identical. The drift measured is the drift nobody controls for.

3.2 Scoring

Unparseable output is scored harmful (fail-closed); fail-open would let a broken decoder report perfect recall. **Unparsed rate is reported per run before any agreement figure**, and a run above 5% is excluded as a generation failure rather than reported with a caveat. No run was excluded: **zero unparsed outputs across all 18,000 evaluations.**

3.3 The statistic, and an honest caveat about it

The primary quantity is **mean pairwise per-item verdict agreement** over all 28 pairs drawn from arm A's 8 runs.

△ **Those 28 pairs are not independent.** They are drawn from a shared set of eight runs, so a naive confidence interval on the mean understates uncertainty. We therefore lead with a statistic that does not have this problem: **the count of items returning identical verdicts across all eight runs simultaneously**, which is a single count over 1,000 independent items.

4. Results

4.1 The floor

998 of 1,000 items returned identical verdicts across all eight runs.

Mean pairwise agreement (28 pairs)	**0.9991**
Worst pair	0.9980
Best pair	**1.0000**
Items ever disagreeing	**2 / 1,000**
Mean pairwise disagreement	**0.09%** = 0.9 verdicts per 1,000

At least one pair of runs was **exactly identical** across all 1,000 items, which bounds the effect from below in a way an average cannot: perfect reproducibility is achievable, just not guaranteed.

4.2 The disagreement is concentrated, not scattered

The two disagreeing items were not different items in different pairs. **Each disagreed in 12 of the 28 pairs**, while the remaining 998 were unanimous everywhere.

This distinction is not cosmetic. A floor of 0.09% spread thinly across nine hundred items would imply pervasive low-level instability. **A floor of 0.09% concentrated in two items that flip roughly half the time implies a small set of genuinely borderline inputs and a stable remainder** — and it is the second, which is why §5's population argument follows.

4.3 Thread count explains nothing (H2 refuted)

	agreement
Arm B, same thread count	0.9993
Arm B, across thread counts	**0.9993**

Identical to four decimal places. The serving stack's own determinism note states that a quantized build is deterministic for a fixed model file, context size **and thread count**, implying thread count is the loose variable. **On this evidence it is not**, and practitioners pinning thread count in the belief that it buys reproducibility are pinning the wrong thing.

4.4 Hardware contributes nothing measurable

	agreement
Within A10G	1.0000
Within L4	0.9980
Across A10G vs L4	0.9990

Cross-hardware agreement (0.9990) is indistinguishable from within-hardware agreement (mean 0.9990). **Two different GPU architectures produce the same verdicts as one architecture produces on repeat.**

4.5 H3 refuted: the floor does not swallow published effects

The comparison target was fixed in advance: our own prior finding that `Q4_K_M` quantization changes **4 to 11 verdicts per thousand** relative to an `f16` reference.

Those effects are 4–12× the floor. They survive. Fixing the target before measuring is what makes this a refutation rather than a rationalization — had the floor come out at 5 per thousand, we would have been obliged to report those findings as inside their own noise.

5. The discrepancy, which is the paper's most useful contribution

5.1 A 10–60× gap

A prior experiment in the same program re-scored two item sets under the same nominal configuration on different containers and observed:

set	items	drift
Qwen3Guard-4B evaded set	97	**1.0%**
Qwen3Guard-0.6B evaded set	105	**5.7%**

Against this paper's **0.09%**. Same model family, same quantization, same seed, same temperature, same serving stack. **The measurements disagree by an order of magnitude or two, and averaging them would be the wrong response.**

5.2 The populations differ, and differ in a way that predicts the gap

The high-drift sets were **not** stratified samples. Every item in them was selected by a two-step filter: it was **caught** by the guard in its original form, and then **missed** after a semantically-preserving rephrasing.

That filter selects for proximity to a decision boundary. An item that flips under a small input perturbation is, by construction, one the model was nearly undecided about. The corpus used in this paper is stratified on the harm label and makes no such selection — most of its items are not close to any boundary, which §4.2 confirms directly: 998 of them never move at all.

5.3 The consequence: a global floor may be the wrong object

If reproducibility depends on item difficulty, then:

- **0.09% is the floor for a stratified corpus** and is the right number for a benchmark reporting aggregate performance over one.
- **It is the wrong number for an effect measured on borderline items** — which is where perturbation studies, adversarial evaluations and fine-grained comparisons concentrate by design.
- **The floor that matters most is the one we did not measure.**

Independent support from prior work. Pinneri and Louizos [GuardMeaning] report paraphrase-induced label flips **highest in the ambiguous score region (0.25–0.75)** and lowest at the confident extremes. That is the same shape our hypothesis predicts, arrived at from a different direction: instability concentrates where the model is undecided. It does not measure a reproducibility floor, but it makes the population-dependence hypothesis considerably more plausible than our two unstable items alone would.

△ **This is still a hypothesis, not a result.** We did not run arm A on a borderline-enriched set. Doing so requires only repeating the protocol on such a set and is the immediate follow-up.

We state it as an open hazard because the alternative — quoting 0.09% as a universal floor — would be exactly the overreach this paper was written to guard against.

5.4 What this means for our own prior work

Applying the finding honestly to results from the same program:

- **Quantization effects (4–11 per 1,000) survive**, at 4–12× the stratified-corpus floor (§4.5).
- **Paired within-run comparisons are unaffected.** The floor is a **between-run** quantity. A McNemar test on discordant pairs within a single run does not inherit it, and conflating the two would understate and overstate different findings simultaneously. The pre-registration forbids blurring this in either direction.
- **Any between-run difference below ~0.1% on a stratified corpus is not reportable**, and any such difference on a borderline-enriched set is currently **unbounded** by evidence.

6. A protocol worth adopting

The floor is cheap to measure and currently nobody reports it.

Minimum viable protocol, one extra run:

1. Run your evaluation **twice**, at identical configuration, in separate processes or containers.
2. Report **per-item agreement** between the two runs alongside your headline metric.
3. **Refuse to claim any between-run difference smaller than the disagreement you just measured.**

Better, eight runs: report the count of items unanimous across all runs, and the identity of the items that are not. That set is diagnostically useful on its own — it is a list of the inputs your guard is genuinely undecided about.

What to report: serving stack and version, quantization, `num_ctx`, seed, temperature, **thread count** (even though §4.3 shows it does not matter here — that is a finding about one stack, and the next stack may differ), and the measured floor.

What not to bother pinning: on this evidence, thread count and GPU architecture. Pinning them is harmless and cheap, but it should not be mistaken for having addressed reproducibility.

7. Threats to validity

One model, one quantization, one serving stack. Qwen3Guard-4B at `Q4_K_M` through Ollama/llama.cpp. A floor that varies by stack is itself a finding and is untested here. vLLM and transformers, with different batching and kernel-selection behavior, may differ substantially.

The population limitation is the paper's own headline hazard (§5) and is not mitigated elsewhere.

Eight runs is a small sample for a rare event. With 2 disagreeing items, the estimate of *which* items are unstable is far less precise than the estimate of *how many*. A larger repetition count would sharpen the disagreement set without much changing the rate.

Non-independent pairs. §3.3. We lead with the unanimity count for this reason, and report the pairwise mean as a secondary figure.

Container assignment is not controlled. Arm A's runs landed on whatever hardware the scheduler provided. Arm C suggests hardware does not matter, but arm A is not a controlled test of that.

We cannot rule out that the two unstable items are corpus artifacts — malformed text, unusual tokenization — rather than genuinely borderline content. Inspecting them is a next step and would sharpen §5.

8. Conclusion

We measured how much a guard evaluation moves when you simply run it again. On a stratified corpus the answer is: **very little.** 998 of 1,000 items are unanimous across eight independent runs, the mean pairwise disagreement is 0.09%, and at least one pair of runs is exactly identical.

That is good news, and it refutes the claim this paper set out to make. It also refutes the standard explanation for guard nondeterminism: thread count, the variable the serving documentation implicates, explains none of the observed drift, and neither does GPU architecture.

What survives is narrower and more useful than the claim we expected to make. The disagreement is concentrated in a tiny number of borderline items rather than spread across the corpus, and a prior measurement on a set composed *entirely* of borderline items showed drift ten to sixty times higher. If reproducibility tracks item difficulty, the floor is not one number, and the number that matters most — the floor where effects are actually measured — is still unmeasured.

Reporting a floor costs one extra run. Until it becomes normal to report one, the field is comparing guards without knowing what a difference is worth.

Open work: the borderline-item floor (§5.3); cross-stack floors for vLLM and transformers; inspection of the two unstable items; and whether the floor scales with model size, since the prior 0.6B observation was five times the 4B's.

References

[**WildGuard**] *WildGuard: Open One-Stop Moderation Tools for Safety Risks, Jailbreaks, and Refusals of LLMs.* University of Washington, Allen Institute for AI, Seoul National University. arXiv:2406.18495, 2024. — source of the evaluation corpus.

[**GuardMeaning**] C. Pinneri and C. Louizos. *Guarding the Meaning: Self-Supervised Training for Semantic Robustness in Guard Models.* Qualcomm AI Research. arXiv:2511.10665, 2025. — establishes that guard verdicts move under meaning-preserving paraphrase; this paper measures how much they move when *nothing* changes, which is the complementary quantity.

[**GuardBench**] *Benchmarking Open-Source Safety Guard Models: A Comprehensive Evaluation.* arXiv:2605.28830, 2026. — an example of the comparison genre whose noise floor is unreported.

[**QuantTemp**] *The Joint Effect of Quantization and Sampling Temperature on LLM Safety Alignment: A Factorial Analysis.* arXiv:2606.29581, 2026. — varies serving parameters deliberately; we hold them fixed and measure the residual.

[**LongGuard**] Z. Chen, X. Wu and S. Hu. *LongGuard: Mechanistic Analysis and Training-Free Mitigation of Long-Context Failure in Safety Guardrails.* arXiv:2608.27580, 2026. > △ **Scope of the prior-art check, stated honestly.** The claim in §1 and §2 that we found no prior > measurement of *repeated-run verdict agreement for guard classifiers* rests on a keyword search of > the guard-evaluation and quantization literature conducted on 2026-08-31, **after** this > experiment ran. It is not a systematic survey. The systems literature on serving nondeterminism is > substantial and we do not claim novelty on the existence of the phenomenon — only on measuring its > magnitude for a safety decision, and on the population-dependence hazard in §5. ---